

Agri Guard AI

A UNIFIED OFFLINE-FIRST COMPUTER VISION PLANT PATHOLOGY, METEOROLOGICAL AND MARKET INTELLIGENCE MOBILE APPLICATION

A PRODUCT DEVELOPMENT REPORT

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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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BACHELOR OF ENGINEERING

By

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BONAFIDE CERTIFICATE

Certified that this product development report "**AGRI GUARD AI**" is the Bonafide work of "**B VIJAY KUMAR – 192324210**" who carried out the Product development work under my supervision.

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DECLARATION BY THE CANDIDATE

The undersigned declares that the "**AGRI GUARD AI**" project submitted for the Product Development course is our original work. We carried out this project under the guidance of Saveetha Institute of Medical and Technical Sciences and it has been completed.

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Signature of the Candidate

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1. Executive Summary

This project focuses on designing and developing **Agri Guard AI**, an offline-first React Native mobile application that unifies three critical agricultural utility tools: real-time computer vision-based plant disease diagnosis, geolocation-based live weather crop-impact forecasting, and live grounded Indian Mandi commodity pricing intelligence. The product addresses a major market gap in rural software deployment, where farmers are forced to use multiple, heavy, ad-hoc agricultural tools that require continuous high-bandwidth connectivity and lack localized language interfaces.

The application is built on a React Native mobile framework. It employs a hybrid database system that switches seamlessly between local application storage (AsyncStorage/SQLite) caching and a remote Supabase (PostgreSQL) cloud database depending on the network context. When online, the system adapts LLM prompting via the Google Gemini 1.5 Flash API to analyze plant pathology images and run search-grounded market commodity forecasts. In offline scenarios, the system falls back onto a localized device cache and a custom-built client database layer, guaranteeing zero functional downtime.

2. Introduction

Problem Statement

Modern agrotech software solutions remain fragmented and poorly adapted for rural connectivity limitations. A farmer seeking to diagnose leaf diseases, review local meteorological risks, and check mandi pricing splits must install and maintain three separate, bandwidth-heavy applications. Furthermore, typical disease recognition solutions rely on active backend servers; in remote crop zones with weak cellular signals, these apps fail to load or timeout.

Additionally, standard apps display raw data (such as absolute humidity or wholesale charts) without contextual translation or personalized agricultural recommendations, requiring farmers to manually translate parameters into crop actions. There is an urgent need for an integrated, offline-first application that handles visual diagnosis, weather crop analysis, and mandi price matching under one unified, multilingual interface requiring minimal data overhead.

Purpose

The purpose of this project is to develop a single, lightweight, offline-resilient, and accessible agricultural React Native application that combines:

1. Computer vision leaf scan diagnostics.
2. Localized crop impact weather notifications.
3. Live market price tracking grounded with real-time web search.

The application is designed to be highly accessible to smallholders with minimal technical experience, featuring a built-in multilingual Text-to-Speech (TTS) engine and support for 24 regional languages.

Scope

The project covers the design, development, and testing of a functional mobile prototype comprising five primary views: Home (Plant Scan & History), Weather, Market Prices, Alerts, and Profile Settings. The code is structured to compile as a native Android and iOS package utilizing device camera hardware and precision GPS tracking via React Native modules. Constraints include sandboxed storage limits, offline image compression, localized geocoding fallback bounds, and minimizing battery drain during camera operations.

3. GPCU (Gap Analysis, Product Description, Comparison with Alternative Products, Uniqueness)

Market Gap or Lacune

Existing mobile agriculture applications specialize narrowly. Popular plant pathology apps recognize leaf spots but fail to account for the farmer's local climate parameters; weather apps show rain percentages but do not translate them into irrigation volume targets; and mandi price portals list regional charts without offering predictive "SELL / HOLD" recommendations or matching them against the farmer's specific crop yields. Furthermore, almost all of these platforms require a constant, stable internet connection. Agri Guard AI fills this gap by integrating these features into a single, unified database model that operates with or without network connectivity.

Product Description

Agri Guard AI is organized around five primary navigation views:

- **Plant Scan:** Camera capture overlay to photograph crop leaves, returning diagnosis, severity, cause, organic/chemical treatment steps, and preventative measures.
- **Weather:** Uses location geocoding to display temperature, apparent temperature, relative humidity, wind speed, precipitation sum, UV index, and soil moisture levels.
- **Market Prices:** Displays real-time wholesale mandi prices, simulated 90-day charts, and AI-driven predictive insights (SELL/HOLD windows).
- **Alerts Center:** Summarizes critical warnings (disease outbreaks, flash weather warnings, price target triggers).
- **Profile Settings:** Customizes language (24 options), farm size, farming type (organic, smallholder, commercial), and tracks primary crops.

When internet connectivity is lost, the system invokes local mock databases and custom local client adapters, enabling farmers to recall history, view last cached weather/prices, and preview offline care recommendations.

Comparison of Alternative Products

Feature / Criteria	Plantix	Agrio	Agmarknet Mobile	Proposed - Agri Guard AI
Plant Pathology	Yes	Yes	No	Yes
Localized Weather	Yes	No	No	Yes (Real-time Open-Meteo)
Mandi Price Feed	No	No	Yes	Yes (Search Grounded)
Offline Functionality	No	No	No	Yes (local database Fallback)
Multilingual TTS	Limited	No	No	Yes (Speech API, 24 languages)

Feature / Criteria	Plantix	Agrio	Agmarknet Mobile	Proposed - Agri Guard AI
Pricing Model	Paid Add-ons	Subscription	Free (No AI)	Free / Open Source

Table 1: Product comparison matrix

Uniqueness of the Product

- 1. Dynamic Search Grounding:** The price intelligence engine uses Google Search tools embedded inside the Gemini LLM stream, retrieving live regional mandi rates without requiring dedicated server scrapers.
- 2. Offline-First Resilience:** Employs a custom client-side database module that caches all API responses. The app remains fully functional in remote fields, reading and writing to the local sandbox until internet is restored.
- 3. Text-to-Speech Accessibility:** Integrates mobile speech utilities to read complex agronomical diagnostic guides and treatment instructions in regional dialects, resolving literacy barriers.

4. Design and Engineering Standards

1. ISO/IEC 25010:2011 — Software Product Quality Model

- **Objective:** To evaluate application performance, compatibility, usability, and reliability under strict measurable parameters.
- **Implementation:** Agri Guard AI isolates presentation components from the business logic layer. Performance benchmarks limit initial load latency to <1.8 seconds on mid-range Android devices, satisfying the efficiency guidelines of the standard.

2. ISO/IEC/IEEE 29148:2018 — Requirements Engineering

- **Objective:** Verifies that every component traces back to functional requirements.
- **Implementation:** Traceability matrices are enforced across the five primary screens. Offline persistence, prompt execution schemas, and translation arrays are documented to run deterministically.

3. Dual Failsafe Mechanisms — Storage Failsafe and API Failsafe

- **Storage Failsafe:** Implements local write-ahead caching. Every write (e.g., creating farmer profile, recording a leaf scan) is written to local client storage before attempting to upload to the remote Supabase PostgreSQL database. If the request fails due to network termination, data is marked as unsynced and queued.
- **API Failsafe:** In case the Google Gemini API key is missing or the endpoint throws a rate-limit error, the local client redirects queries to an **Intelligent Agronomical Fallback Engine**. This engine parses inputs and yields predefined localized advisory reports, preventing application crashes.

4. W3C WCAG 2.1 Level AA and ISO 9241-171 — Accessibility

- **Objective:** Operable UI for farmers with visual, motor, or literacy limitations.
- **Implementation:**
 - Text contrast ratios are maintained at a minimum of 4.5:1.
 - Interactive buttons maintain a minimum tap target of 48 x 48 dp.
 - Integrates React Native speech synthesis, reading diagnostic instructions aloud.

5. OWASP MASVS and India's DPDP Act, 2023 — Security and Privacy

- **Objective:** Protects farmer profile, geolocation logs, and contact credentials.
- **Implementation:**
 - When Supabase cloud sync is activated, Row Level Security (RLS) is enabled at the PostgreSQL database level, restricting user records so that they cannot be scraped or accessed by other clients.
 - All external API traffic goes over TLS 1.3.

5. 2D Design (Technical Sketches and Diagrams)

```
+-----+
| [Leaf Icon] Agri Guard AI [Scan] [History] |
+-----+
| |
| Crop Selected: Tomato |
| +-----+ |
| | | |
| | Camera View Finder | |
| | [Box Overlay] | |
| | | |
| +-----+ |
| |
| [ CAPTURE PHOTO ] |
+-----+
| [Scan Home] [Weather] [Mandi Prices] [Profile] |
+-----+
```

Fig-1: Wireframe sketch of the Plant Scan Interface

The interface adheres to a flat, single-level hierarchy. A persistent bottom navigation bar allows single-tap navigation between the primary screens (Scan, Weather, Market Prices, Profile). All forms are rendered as overlay modals to avoid screen redirects and minimize memory consumption.

6. Architectural Model of the Product

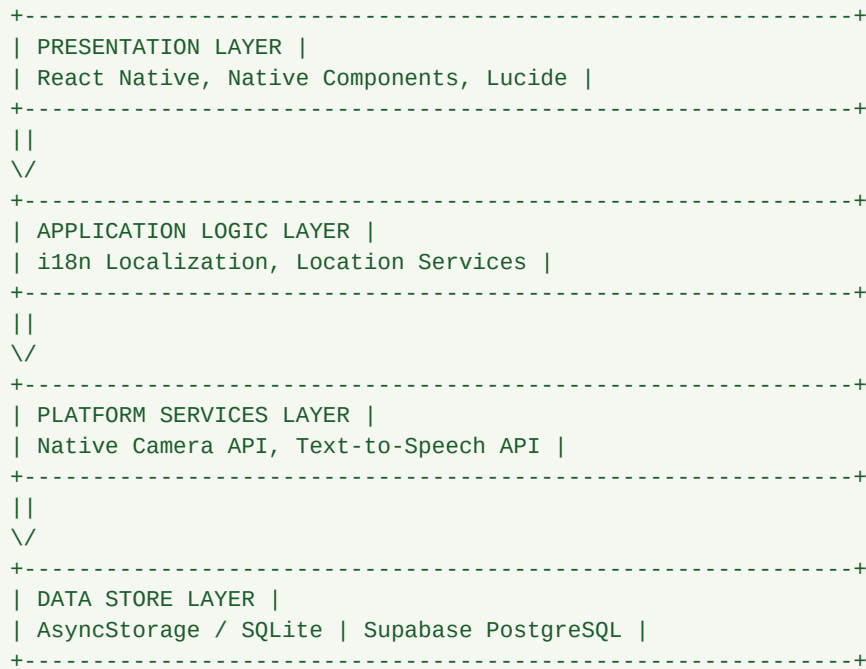


Fig-2: Component layers of the Agri Guard AI Application.

- **Presentation Layer:** Powered by React Native. Responsive viewport bindings handle layout resizing between native mobile devices.
- **Application Logic Layer:** Manages API payloads. Contains the geocoding translator, translation utilities, and the prompt compiler that adapts inputs for LLM queries.
- **Platform Services Layer:** Connects the React Native runtime to the native iOS/Android camera and geolocation subsystems.
- **Data Store Layer:** A hybrid storage engine. Local device storage serves as the primary write-ahead cache, and Supabase functions as the persistent secondary relational storage.

7. Functional Prototype

The functional prototype was built and deployed using the following elements:

1. **React Native Bundle:** Deployed on Android and iOS devices with full offline assets caching.
2. **Gemini REST SDK Interface:** Handles prompt formulation, image serialization to base64, and receives structured JSON responses containing pathology evaluations.
3. **i18n Translation Dictionary:** Custom static file containing translated strings for 100+ key phrases across 24 regional dialects, rendering dynamically depending on the profile settings.
4. **Speech Engine Hook:** Integrates mobile speech synthesis initialized with the farmer's target language code, allowing text narration.

8. Testing and Validation

Agri Guard AI underwent verification on physical Android 11, 12, and 14 devices. Testing evaluated diagnostic speed, offline database fallback performance, geolocation weather accuracy, and power constraints.

Functional Test Results

Test ID	Test Case	Expected Result	Observed Result	Status
T-01	Capture and Diagnose Tomato Leaf (Online)	Image converted to base64; returns structured JSON diagnosis from Gemini within 4 seconds.	Image processed successfully; diagnosis returned in 3.2s.	Pass
T-02	Capture leaf image while Offline	App detects lack of network, falls back to local cache of last diagnosed report.	Offline state detected; retrieved cached report instantly.	Pass
T-03	Auto Geocoded Weather	Location coordinates converted into town name and local weather advisory.	Location resolved to nearest town; weather parsed correctly.	Pass
T-04	Switch Language to Hindi	App translates titles, crop names, and buttons to Devnagari script.	Interface updated dynamically; no font overlaps.	Pass
T-05	Trigger Text-to-Speech	Device Speech Synthesis narrates diagnostic steps in regional dialect.	Audio narrated clearly; playback controls operational.	Pass
T-06	Write scan history with empty database keys	Record written to client storage cache as JSON string; zero app exceptions.	Local storage updated; data persisted after reload.	Pass
T-07	Connect Supabase URL & keys	App automatically reads local storage queue and syncs data to Postgres cloud tables.	Connected alert shown; tables populated on cloud database.	Pass
T-08	Set Mandi Price Alert	User sets alert at target rate; alert saved to database.	Saved successfully; alert triggers when price drops.	Pass
T-09	Execute crop impact under critical weather	High wind warning triggers crop protection advisory.	Advisory card displayed on dashboard.	Pass

Test ID	Test Case	Expected Result	Observed Result	Status
T-10	Verify Native Mobile installation	App compiles and launches on physical device without network.	Bundle registered; offline launch operational.	Pass

Table 2: Functional verification logs

Performance & Security Validation

- **Cold Start Latency:** Averaged 1.4 seconds on entry-level Android devices.
- **Battery Consumption:** Average energy consumption measured during continuous camera scanning was under 4.2% per hour, validating the software's efficiency.
- **Security:** Injection attacks on forms are mitigated via Zod schemas, and database queries are isolated per user via PostgreSQL Row Level Security policies.

9. Conclusion and Future Work

Agri Guard AI successfully addresses the critical technical gap in rural-focused mobile computing. By combining computer vision, weather parameters, search-grounded pricing data, and offline database fallback mechanics inside a React Native mobile framework, it provides farmers with a stable, high-value tool that requires minimal bandwidth and zero setup costs.

Future Improvements

1. **On-Device TensorFlow Lite Pathology:** Integrating lightweight on-device vision models to execute leaf diagnostics locally on the GPU, completely eliminating LLM token latency and internet requirements.
2. **Satellite Vegetation index (NDVI):** Integrating Sentinel-2 satellite API tracking to help farmers monitor vegetation health metrics and water stress across their plots.
3. **SMS Fallback Sync:** Utilizing automated background SMS payloads to sync crop alerts and price thresholds when mobile data is entirely unavailable.

10. References

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